香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen
人工智能學院
School of Artificial Intelligence

AIE1901-S1 Lifelong AI:

Lecture 4: “What to Transfer” in Model-based Transfer Learning

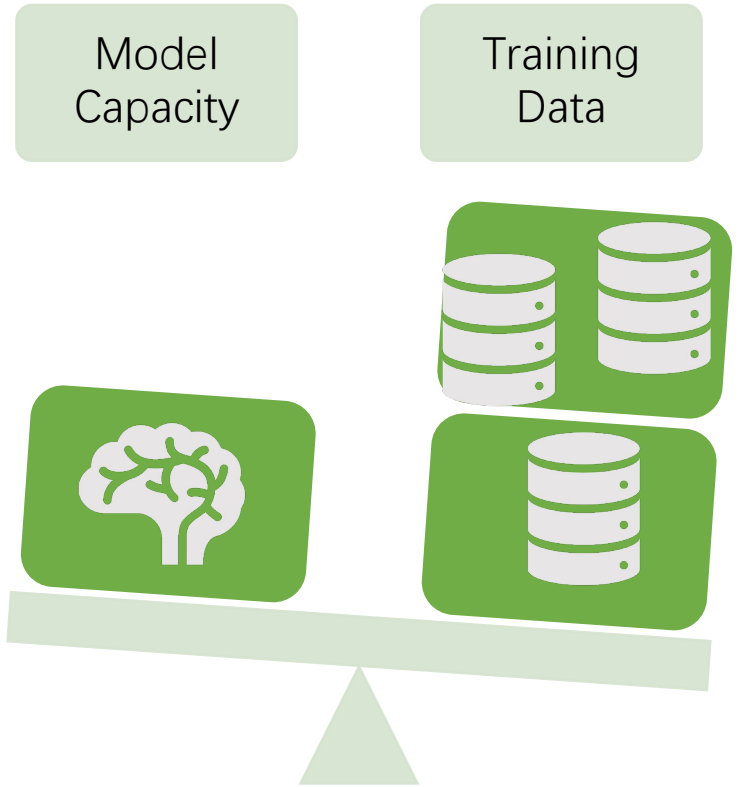
Yang Li

2026/10/12

Agenda

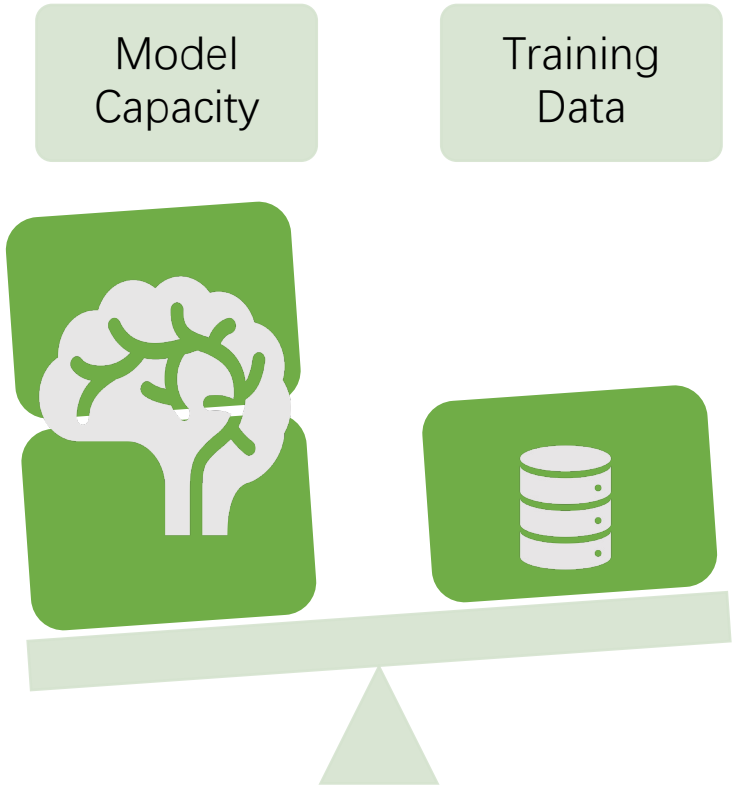
- Review: No Free Lunch Hypothesis & Transfer Learning
- *What to Transfer in Deep Neural Network*

Review: No Free Lunch Theorem



- **Good**: fast to train and predict, small storage
- **Bad**: capacity is limited, unable to learn difficult task (underfit)

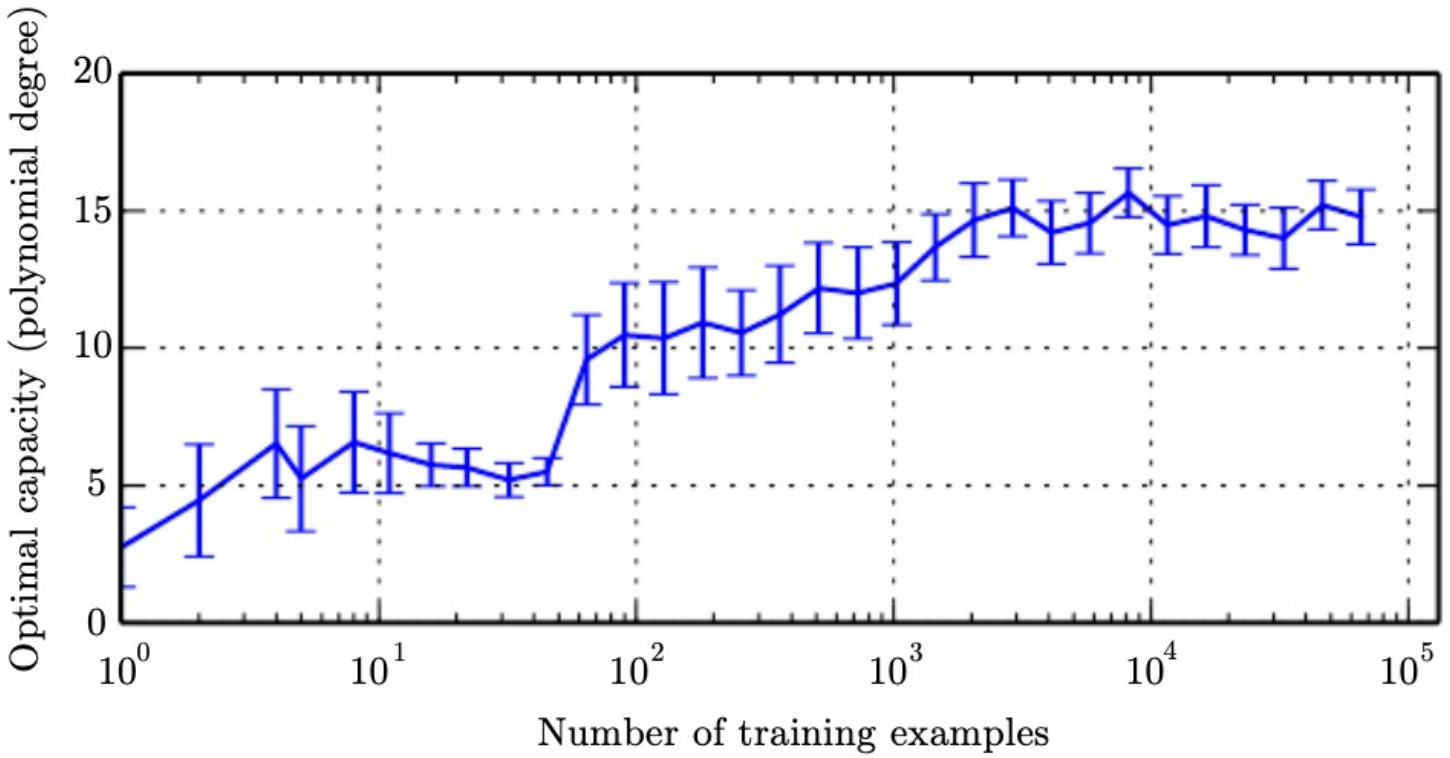
No Free Lunch Theorem

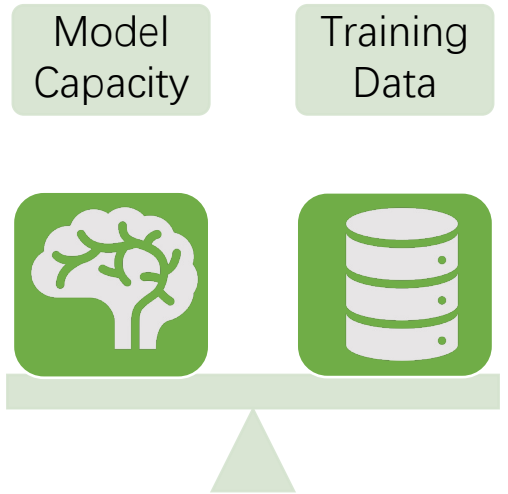


- **Good**: model is capable of learning
- **Bad**: model can be sensitive to training data sampling (overfitting), larger model makes it slow to train and predict

No Free Lunch Theorem

- No machine learning algorithm is universally better than another (Wolpert 1996)



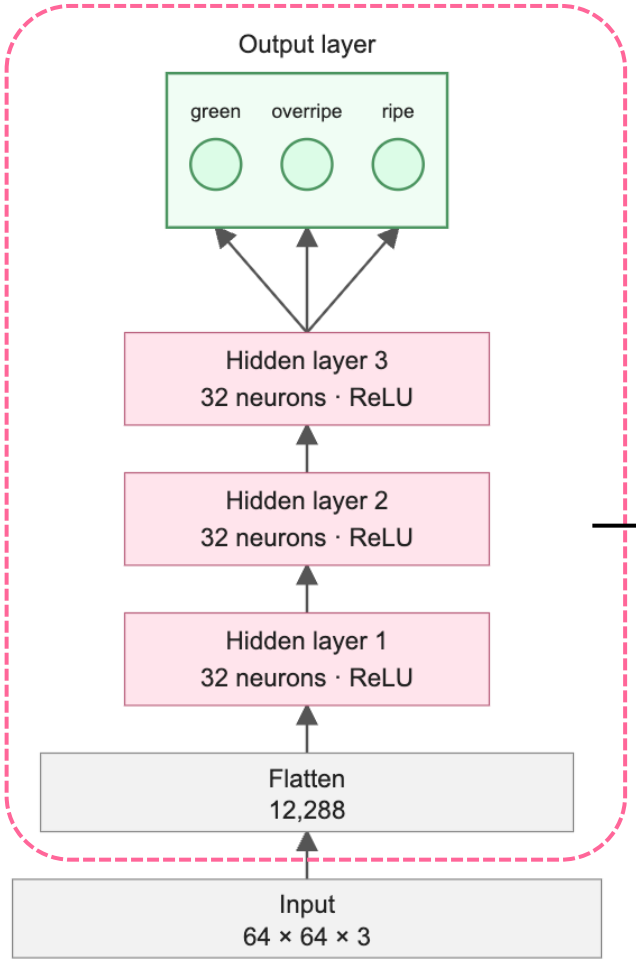


The **optimal model capacity** depends on the amount of data and the task complexity

Review: Model Finetuning

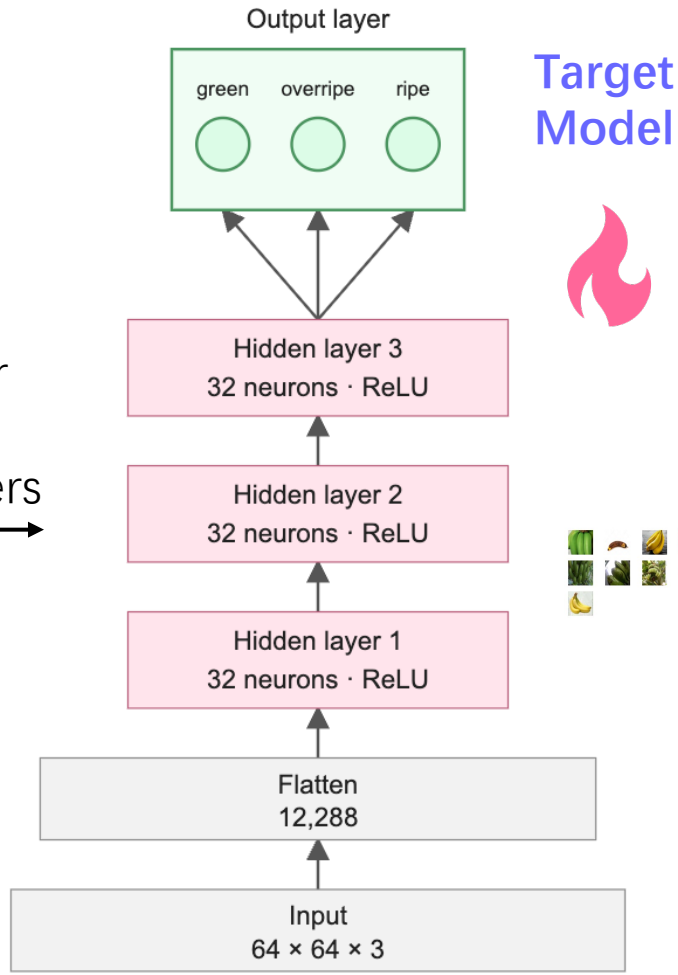
Source Model





Transfer model parameters

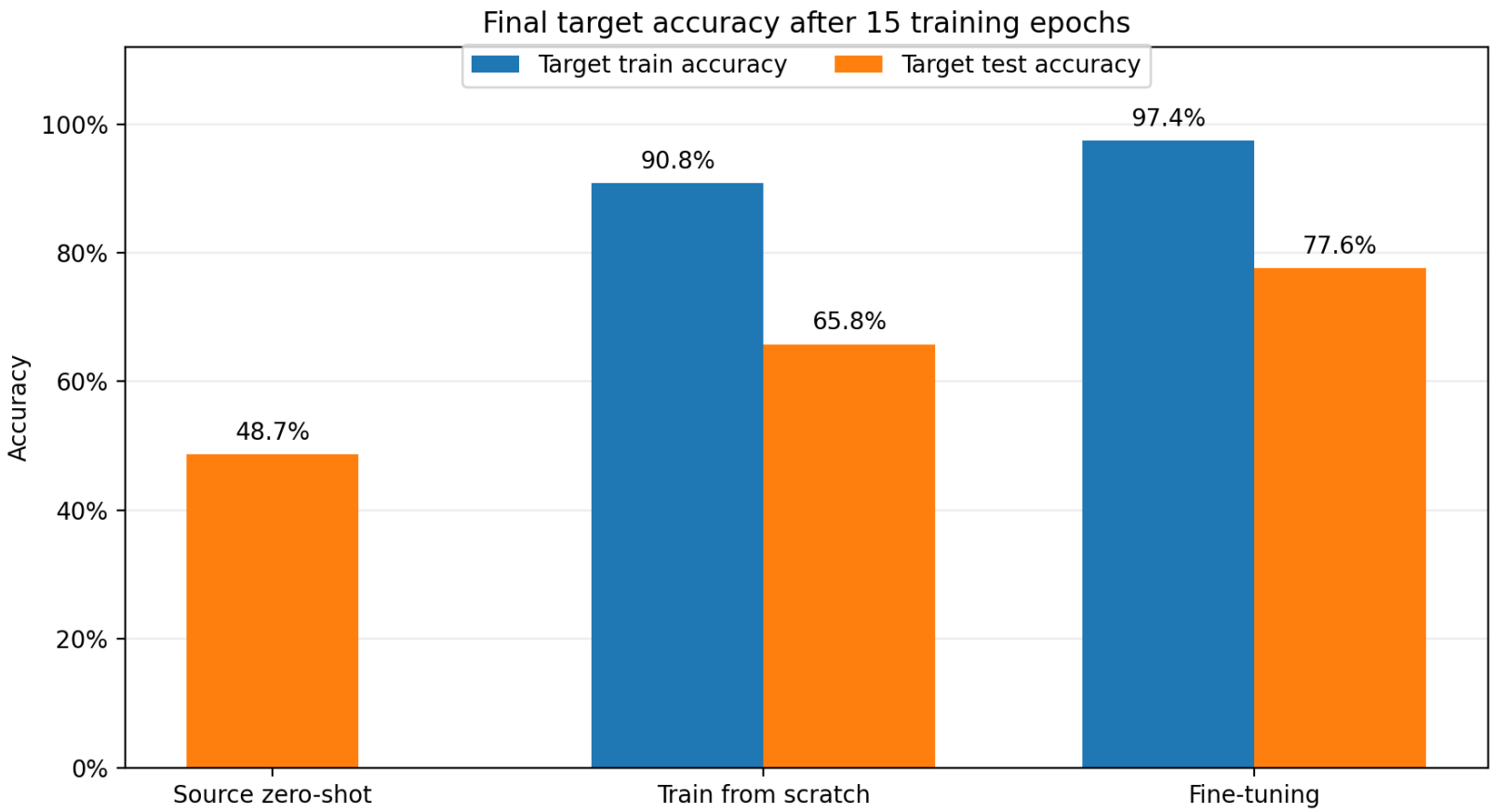
Target Model



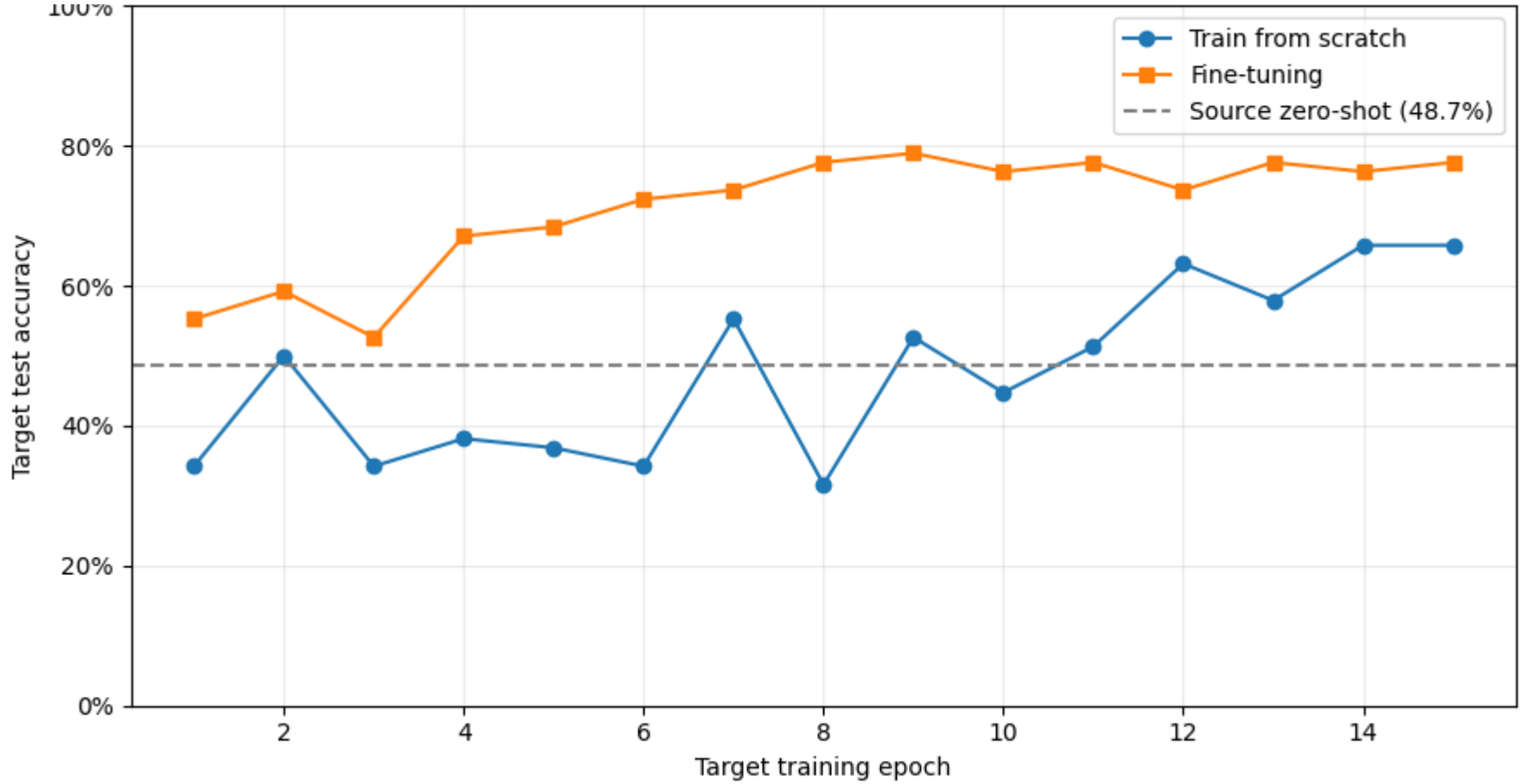


- **Step 1**: Initialize target model feature extractor with source model parameters
- **Step 2**: Continual update model parameters using target training data

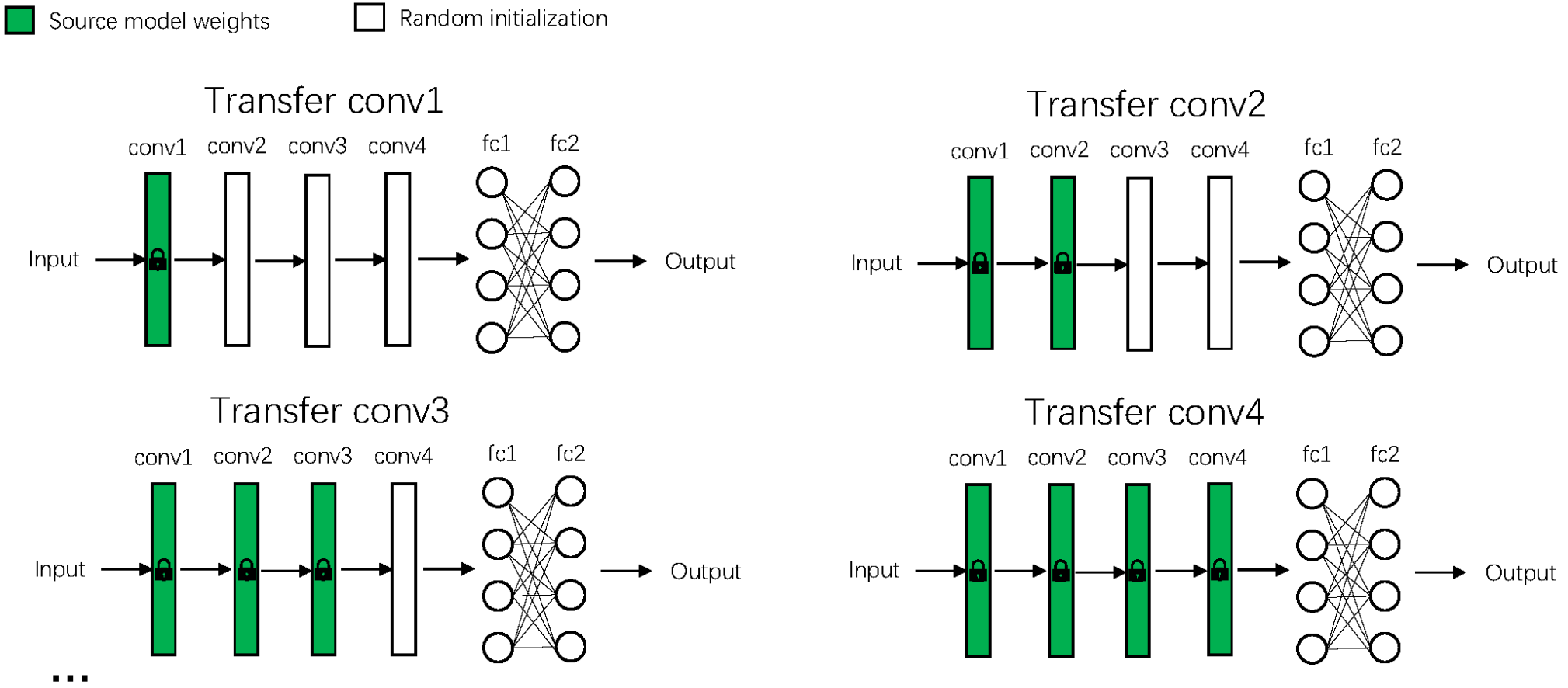
Finetuning vs. Training from Scratch



Finetuning vs. Training from Scratch

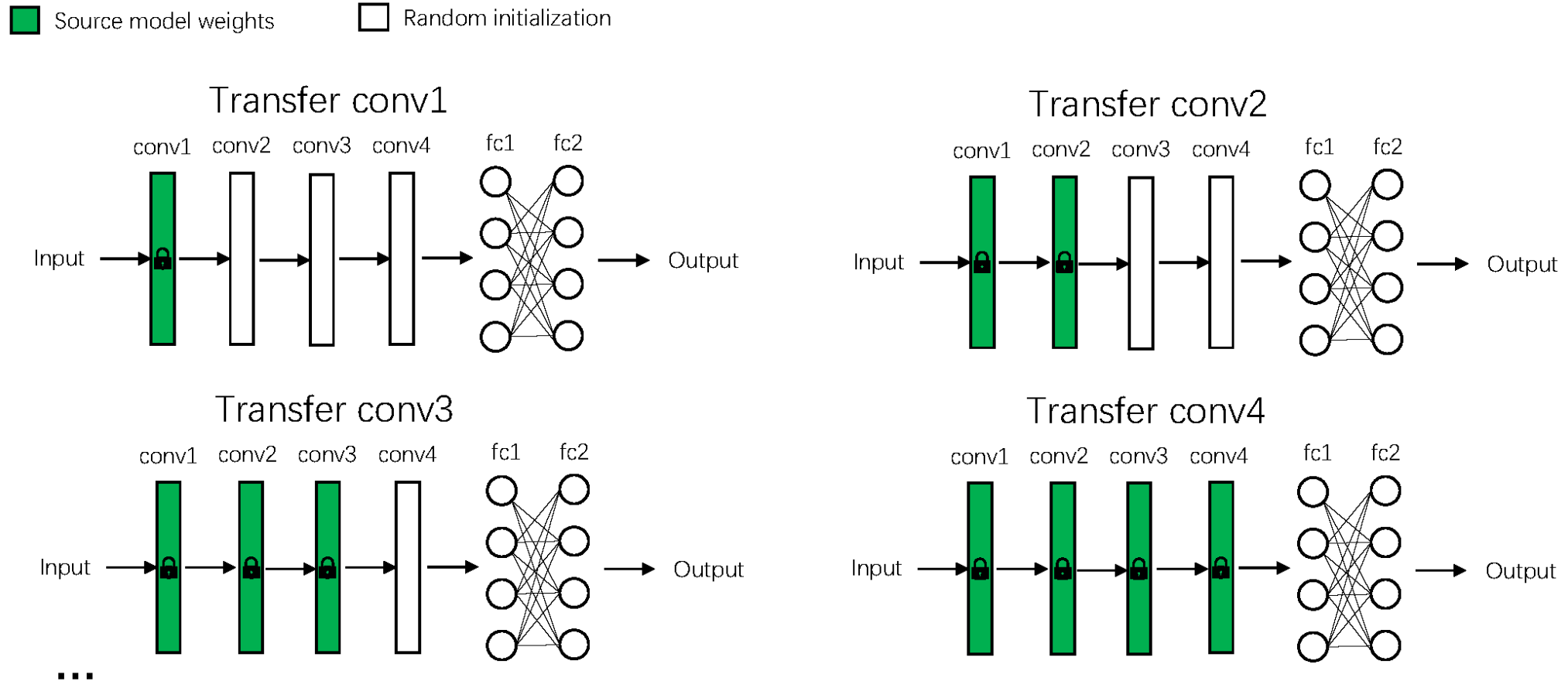


Finetuning Selected Neural Network Layers



- **Step 1**: Initialize target model feature extractor with source model parameters
- **Step 2**: Continual update model parameters using target training data

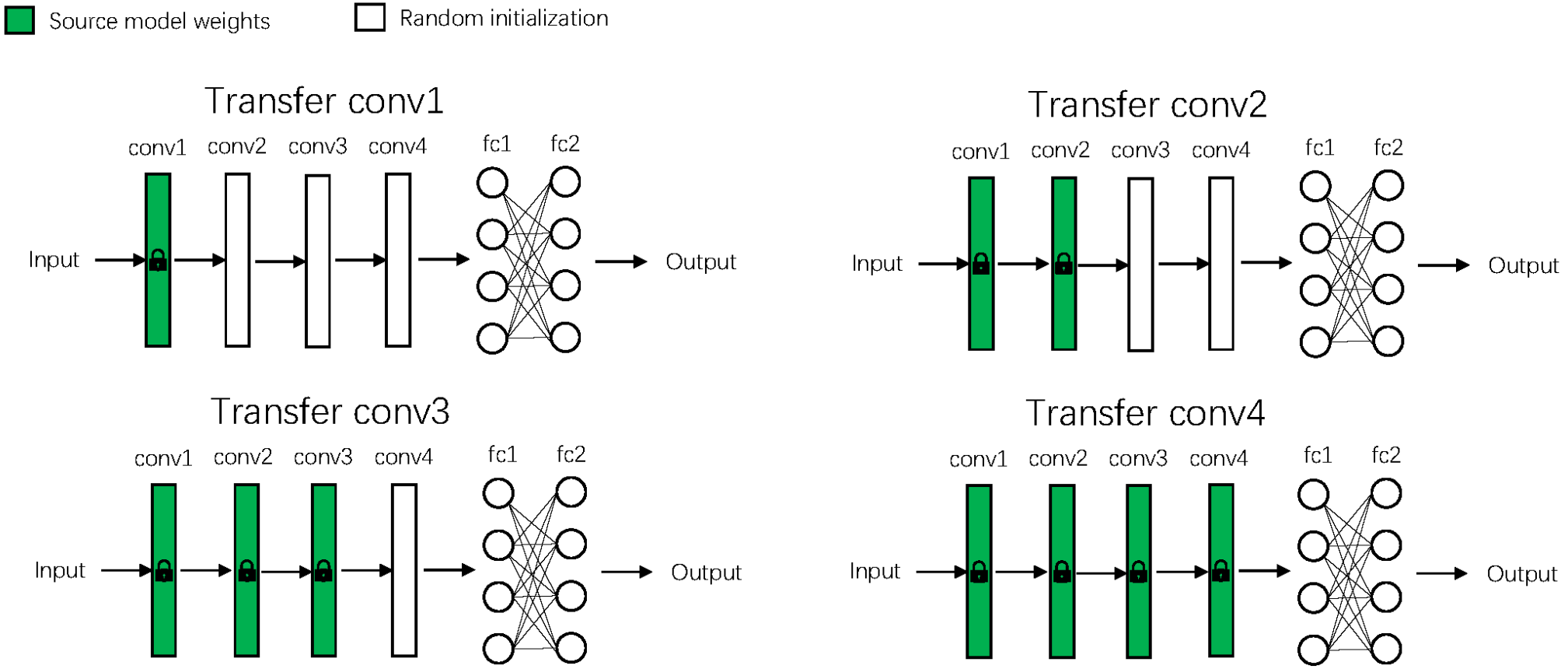
Finetuning Selected Neural Network Layers



- **No Free Lunch Theorem in Transfer Learning**: What-to-transfer also depends on the amount of target data and task!

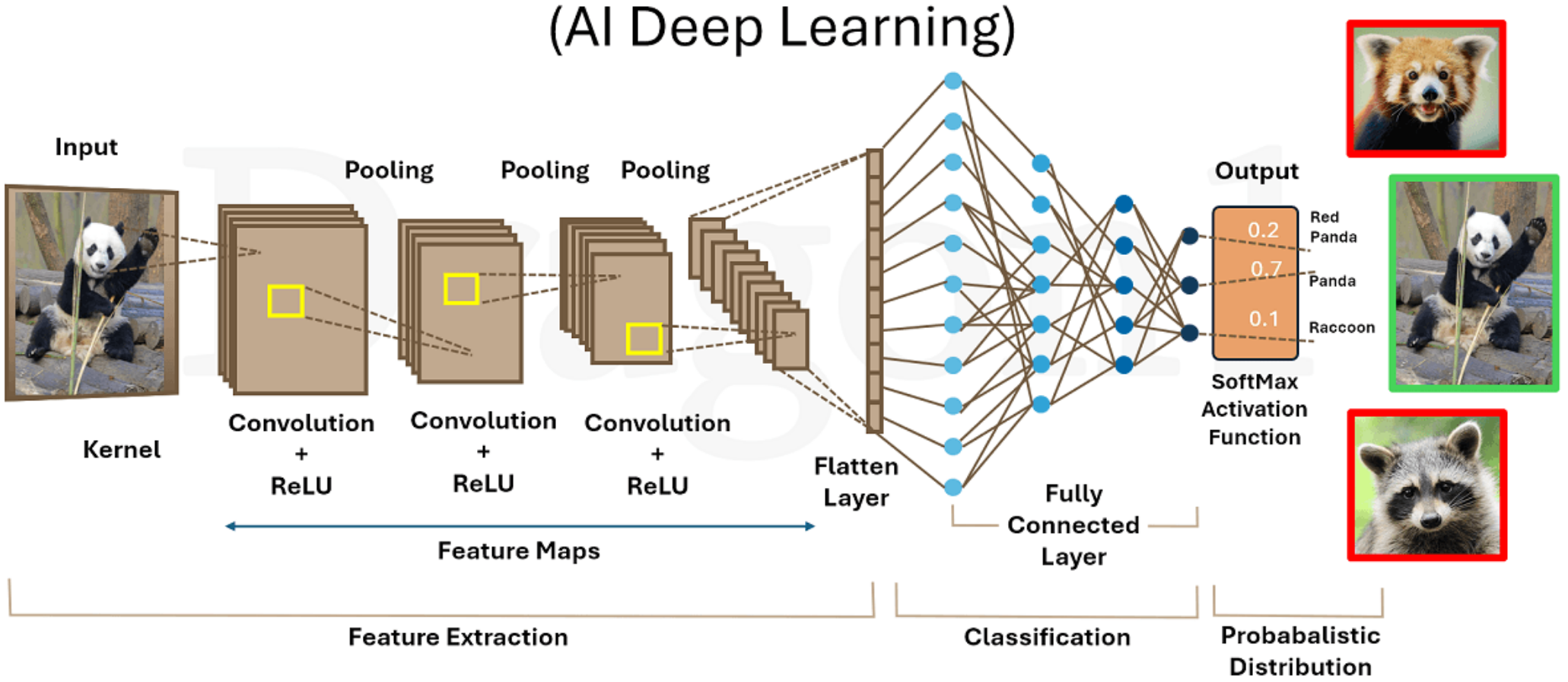
No Free Lunch Theorem in Transfer Learning

- What-to-transfer also depends on the amount of target data and task!



What layers should be frozen and what layers should be updated?

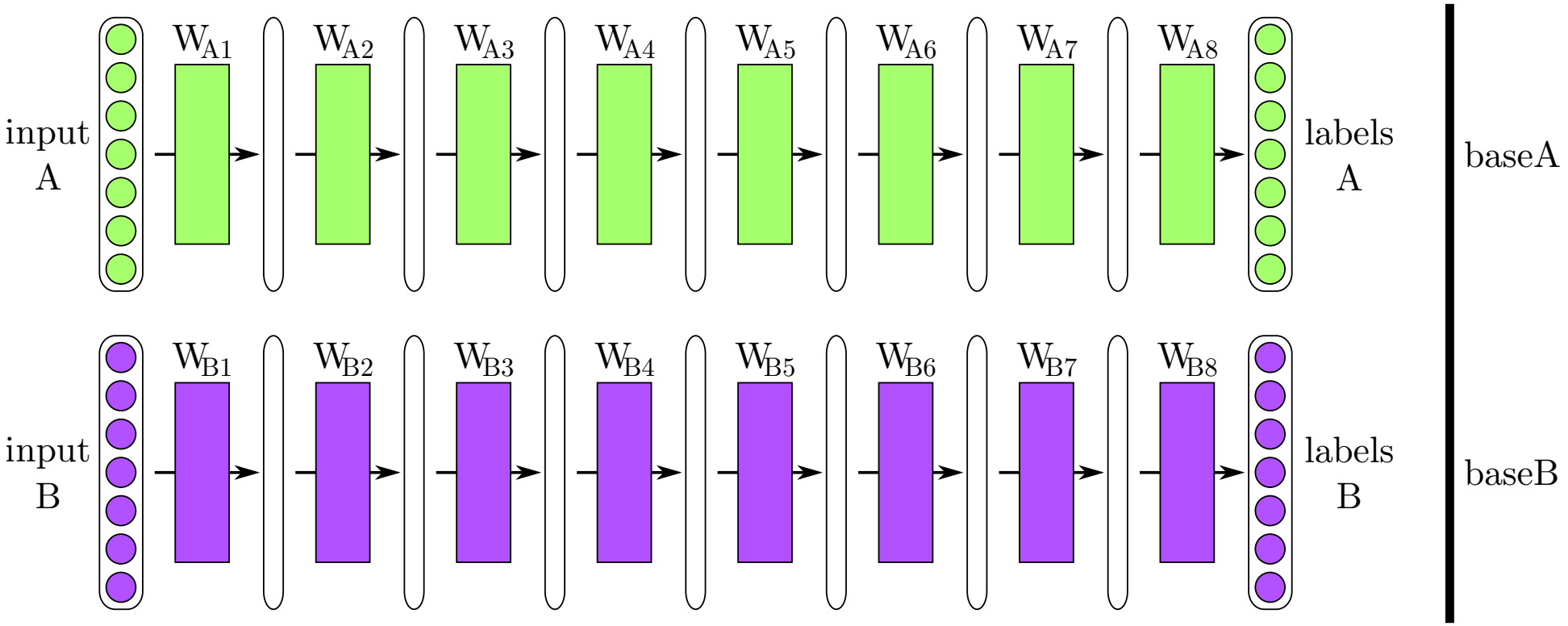
Background: Convolutional Neural Networks (CNN) for Image Classification



© Copyright 2025, Dragon1, www.dragon1.com

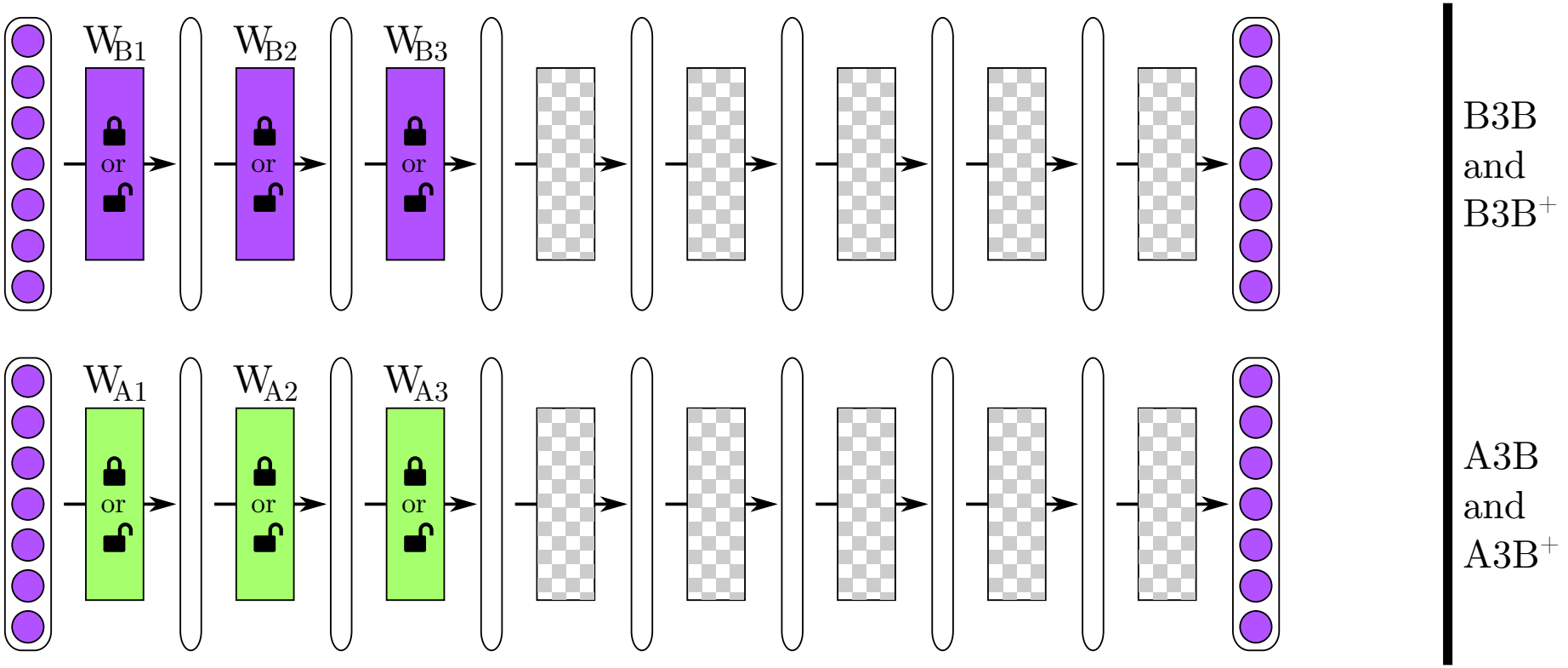
CNN introduction video https://www.youtube.com/watch?v=n4lgQEmEUJM&t=55s

How Transferable are Features in Neural Network?



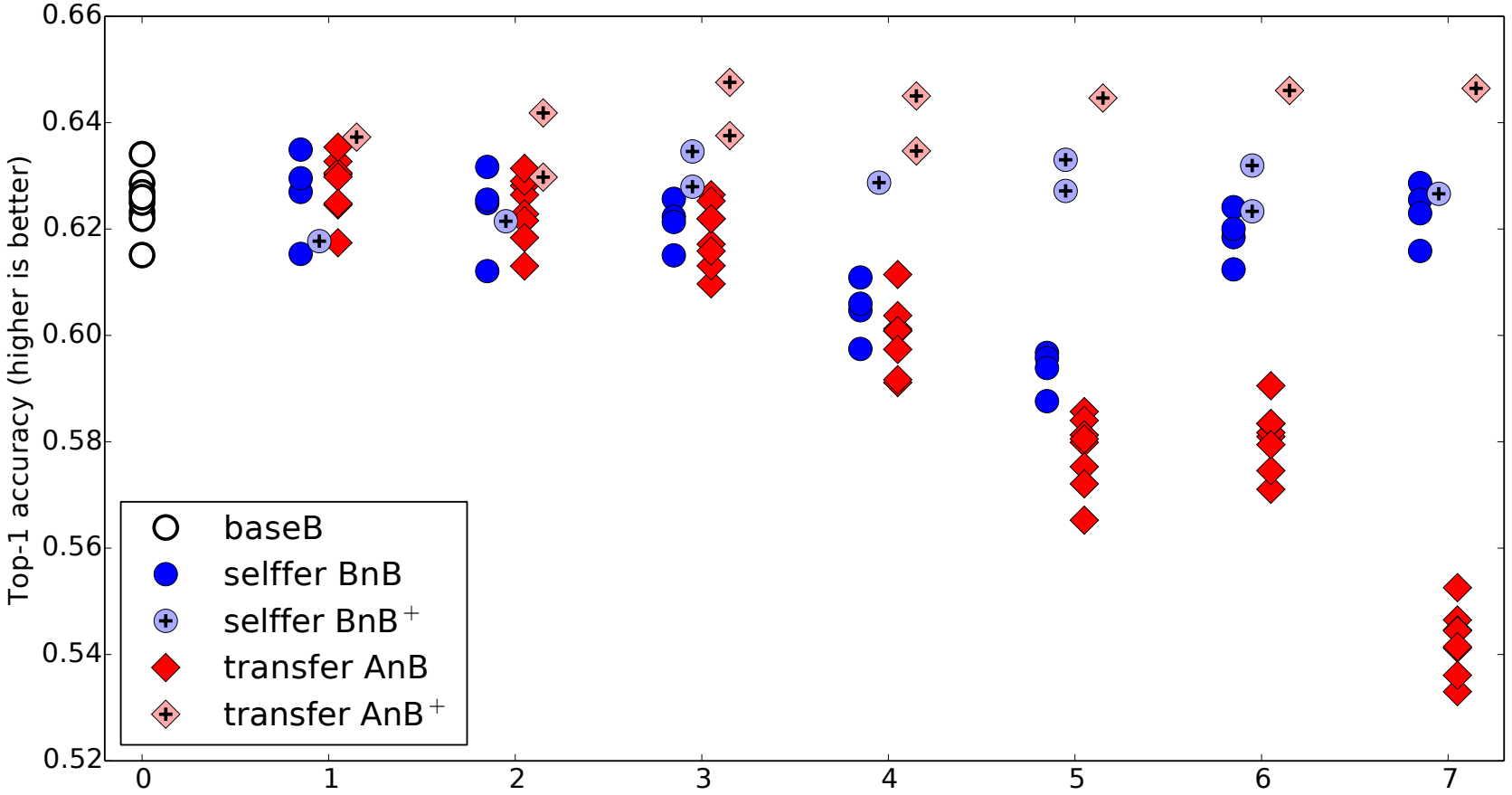
Yosinski, J., Clune, J., Bengio, Y., & Lipson, H.. **How transferable are features in deep neural networks?** Advances in Neural Information Processing Systems (NeurIPS) 2014

How Transferable are Features in Neural Network?



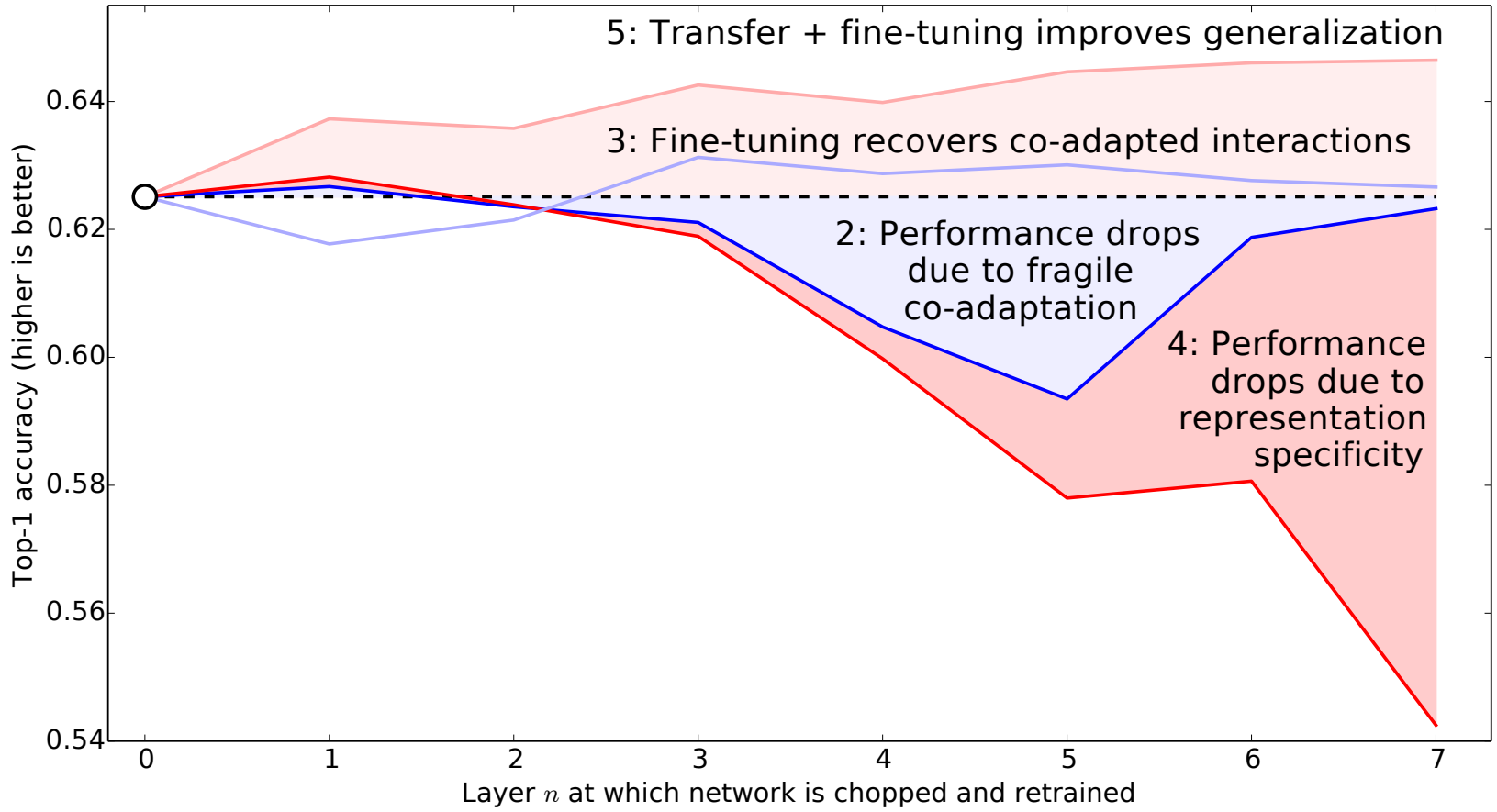
How Transferable are Features in Neural Network?

- Transferring **higher-level features** increasingly reduces accuracy, while fine-tuning consistently improves performance.

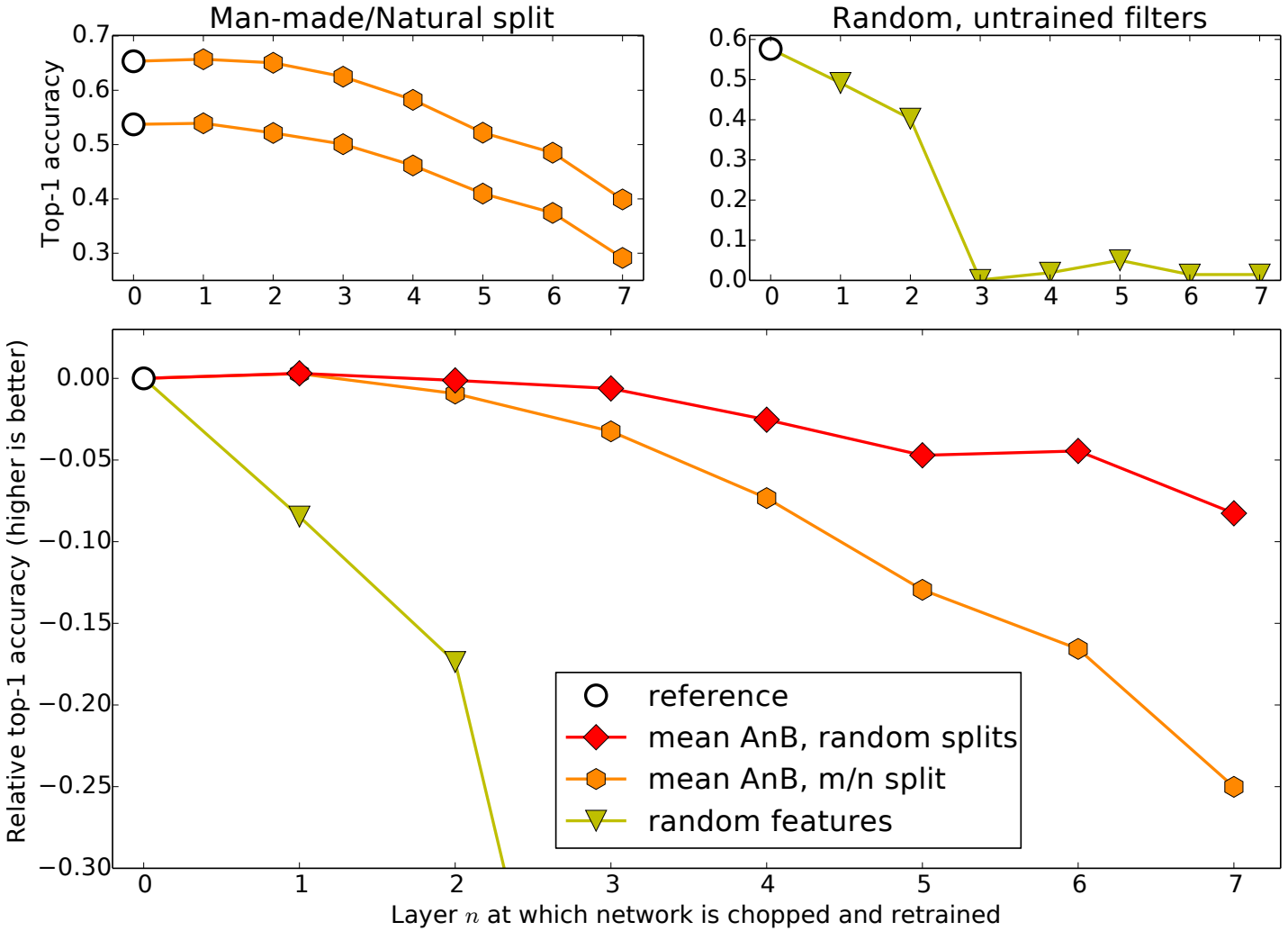


How Transferable are Features in Neural Network?

- Performance drops due to fragile **feature co-adaptation** and **increasing task-specificity**;
- fine-tuning helps recover lost accuracy.



How Transferable are Features in Neural Network?



Practical Constraints

In practical model application scenarios, we are also concerned with

- Data collection cost
- Training Efficiency
- Source model cost
- Quality of the source model

Next lecture: parameter-efficient transfer learning